Alyssa Johnson

Experimental physicist building scalable ML infrastructure for forecasting, signal detection, and quantitative modeling.
U.S Citizen | San Diego, CA | alj010@ucsd.edu | c: +1 240 678 8942 | LinkedIn | github.io | *References available upon request*

SKILLS

Quantitative Finance: Options Pricing & Greeks, Volatility Modeling, Statistical Inference, Bayesian Modeling, Stochastic Processes, Signal Processing, A/B Testing, Anomaly Detection, Bloomberg Terminal & OpenBB
ML / Stats: Time Series Forecasting, Tree-Based & Ensemble Models (XGBoost, LightGBM), Deep Learning, NLP/LLMs, RAG Systems, Vector Embeddings, Tensor Flow, PyTorch, Hugging Face, Pinecone, Weights and Biases
Programming & Systems: Python (Pandas, NumPy, scikit-learn), C++, SQL, Linux, Git, Distributed Computing, Spark/PySpark, PostgreSQL, Redis, Airflow, Parquet, Flash, APIs
Cloud, Infrastructure & DevOps: AWS, GCP, Docker, Kubernetes, OpenShift, CI/CD, Grafana

EXPERIENCE

UCSD Triton Quantitative Trading Club - Lead, Strategy Development Research Group — March 2026 - *present*
- Developed delta-hedged SPY options strategies with vega-targeted sizing, IV regime filters, and automated Greeks-based execution logic; achieved Sharpe ratios of 1.3–1.7 across multi-regime backtests while reducing drawdowns through dynamic hedging. 🌐

UCSD Optical & Infrared Laboratory - Graduate Student Researcher — April 2025 - *present*
- **Award** → DoE High Energy Physics Consortium for Advanced Training Fellowship → *fellowship is ongoing*
- **Award** → University of California, San Diego Sloan Scholars Fellowship → *fellowship is ongoing*
- Built QPulse, a multi-asset signal generation platform ingesting SEC filings, financial news, and market data using distributed ETL pipelines, Redis caching, and RAG-based LLM workflows on AWS/Docker/Kubernetes; processed 50k+ financial events daily while reducing retrieval latency by ~42%.
- Constructed an in-database ML forecasting pipeline using MySQL HeatWave and vectorized query execution, reducing inference latency by 38% relative to external Python baselines across financial forecasting workdls. 🌐
- (Dissertation) Assembled and validated SiPM detector systems for nanosecond-scale optical transient detection; implemented parallelized signal-processing workflows reducing analysis runtime by ~60% across high-dim. datasets. 🌐

UCSD Cosmology Group - Graduate Student Researcher — August 2022 - April 2025
- **Award** → GEM Fellowship Program
- (M.S. Thesis) Studied detector non-idealities and simulated detector responses; developed stochastic signal models to quantify systematic effects on SAT-1 science products (co-authored SPIE 2024). 🌐

NASA Goddard Space Flight Center - NASA GSFC Intern (x3) — Aug. 2021 - Aug. 2022
- **Awards** → NASA Sally Ride Scholarship, SGAC – NASA Scan Scholarship
- Main Project: Delivered single-user relay and handover experiments that demonstrated user data services for optical communications, defining scheduling parameters and spacecraft configurations to support LCRD mission operations.

Cal Poly, Humboldt, Gravitational Physics Lab - Student Researcher — February 2019 - July 2022
- **Award** → Cal-Bridge Program Scholarship
- (Senior Thesis) Optimized an active servo-leveling control system for a precision short-range gravity experiment using numerical modeling and sensor feedback analysis; published in J. Undergrad. Rep. Phys. (2023). 🌐

Montana State University Solar Physics REU - Student Researcher 🌐 — June - August 2021

EDUCATION

- University of California, San Diego | **PhD Candidate in Physics** — *Expected Spring 2028*
- University of California, San Diego | **MSc in Physics** — grad. June 2025
- California State Polytechnic University, Humboldt | **BSc in Physics, *Cum Laude*** — grad. Dec. 2021

PUBLICATIONS

- **Johnson, A**. et al. *Optimization of an Active Leveling Scheme for a Short-Range Gravity Experiment.*" J. Undergrad. Rep. Phys. 33, 100002 (2023)
- Bhimani, Sanah, et al. (incl. **Johnson. A.**). "*The Simons Observatory: Deployment and current configuration of the Observatory Control System for SAT-MF1 and data access software systems.*" Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII. Vol. 13102. SPIE, 2024.